

```
1 `timescale 1ns / 1ps
2 module MelodyScale(
3     input CLK,
4     input [7:0] BTN,
5     output PWM
6 );
7     parameter DO_PERIOD = (125000000/262)-1;
8     parameter RE_PERIOD = (125000000/294)-1;
9     parameter MI_PERIOD = (125000000/330)-1;
10    parameter PA_PERIOD = (125000000/349)-1;
11    parameter SOL_PERIOD = (125000000/391)-1;
12    parameter RA_PERIOD = (125000000/440)-1;
13    parameter SI_PERIOD = (125000000/494)-1;
14    parameter DOS_PERIOD = (125000000/523)-1;
15    reg [31:0] top_in;
16    reg [31:0] cmp_in;
17    TimerCounter TimerCounter_0(CLK, top_in, cmp_in, PWM );
18    always @(BTN) begin
19        if(BTN==128) begin
20            top_in = DOS_PERIOD; cmp_in = DOS_PERIOD/2;
21        end
22        else if(BTN==64) begin
23            top_in = SI_PERIOD; cmp_in = SI_PERIOD/2;
24        end
25        else if(BTN==32) begin
26            top_in = RA_PERIOD; cmp_in = RA_PERIOD/2;
27        end
28        else if(BTN==16) begin
29            top_in = SOL_PERIOD; cmp_in = SOL_PERIOD/2;
30        end
31        else if(BTN==8) begin
32            top_in = PA_PERIOD; cmp_in = PA_PERIOD/2;
33        end
34        else if(BTN==4) begin
35            top_in = MI_PERIOD; cmp_in = MI_PERIOD/2;
36        end
37        else if(BTN==2) begin
38            top_in = RE_PERIOD; cmp_in = RE_PERIOD/2;
39        end
40        else if(BTN==1) begin
41            top_in = DO_PERIOD; cmp_in = DO_PERIOD/2;
42        end
43        else begin
44            top_in = 0; cmp_in = 0;
45        end
46    end
47 endmodule
```